

# Online Resource 1: Proposed Empirical Designs

Supplement to *Grounding Without Corrective Control:  
Truth-Tracking Profiles for Large Language Models*

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## 1 PURPOSE

This resource specifies three component tests proposed in the accompanying article: a crossed route-by-task design, a comparison that varies checking-route independence, and a temporal test of live answerability. The designs don't jointly distinguish the decomposition from an unrestricted effective-coverage description. Their purpose is to test whether a decomposition fixed before the outcomes are observed supports selective transfer, diagnosis, and intervention. No empirical results are reported; the numerical results below come only from constructed data.

Table 1 states the inferential role of each design. It treats the tests as validation arguments: an observed contrast supports a specified component only under the declared architectural boundary, auxiliary assumptions, and manipulation checks. None is a crucial experiment against every informational redescription.

## 2 FAKE-DATA CHECKS OF EXPLANATORY SURPLUS

Constructed data can test whether a proposed model and design recover a known structure and can reveal cases in which a simpler comparator is sufficient (Gelman et al., 2020). They aren't evidence that the decomposition is true. The exercise here instead asks whether the declared comparison is capable of exposing both success and failure before empirical outcomes are observed.

Let  $p_0$  be an arrangement's probability of an initially correct answer. For a correction opportunity, let  $x$ ,  $d$ , and  $i$  represent target access, detectability, and checking-route independence as operationalized by task-specific conditional non-overlap, and let  $c = xdi$  be the route's conditional signal. Let  $a$  and  $u$  represent attribution and uptake,  $f$  the false-signal probability on an initially correct answer,  $\rho$  the probability that an exercised correction succeeds, and  $\eta$  the probability that an unwarranted

| Test             | Decomposition                                                                                                 | Strongest coverage reply                                                                                          | Evidential role                                                                          |
|------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Route by task    | A supplied route should help most where the task requires it.                                                 | Conditional or effective coverage predicts the same pattern by distinguishing kinds of task-relevant information. | Tests the proposed route carving and selective transfer, not uniqueness.                 |
| Checker relation | Lower conditional error overlap should improve repair when marginal accuracy and baseline inputs are matched. | Conditional coverage predicts the advantage by representing information added given generator error.              | Tests checking-route independence; distinguishes only a fixed input-coverage comparator. |
| Temporal update  | A live route should transmit a source update that a frozen record can't.                                      | Dynamic coverage predicts the same result because follow-up information differs.                                  | Operationalizes live answerability; doesn't discriminate against dynamic coverage.       |

Table 1: Predictions and limits of the three component tests.

intervention causes harm. The structural data-generating model is

$$\begin{aligned}
 P(\text{repair} \mid \text{initial error}) &= c\alpha\rho, \\
 P(\text{harm} \mid \text{initially correct}) &= f\alpha u\eta, \\
 P(\text{final correctness}) &= p_0(1 - P(\text{harm})) + (1 - p_0)P(\text{repair}).
 \end{aligned}$$

This multiplication is one deliberately simple data-generating specification, not a profile score or an assumption that the features are mutually independent. It represents a serial path whose stages all have to succeed on a correction opportunity. The fixed conditional-coverage comparator omits  $a$  and  $u$ , using  $c\rho$  and  $f\eta$ ; the input-coverage comparator uses target access alone for repair. Each model estimates one repair and one harm multiplier from training cells. To isolate the route comparison, every model receives the same true  $p_0$ ; an empirical implementation would estimate that baseline with uncertainty.

The simulation draws 30 heterogeneous arrangements, record and measurement tasks, and five equal-cost choices: no intervention, retrieval, measurement, independent checking, and enforced uptake. Each arrangement–task–intervention cell contains 120 trials. Four intervention–task combinations are withheld from fitting. The models are compared on those cells by logarithmic loss for repair conditional on initial error and by regret from choosing an intervention other than the truly best one. Table 2 reports means over 200 replications from a fixed master seed. Negative differences favour the decomposition; positive differences favour conditional coverage.

The first world verifies that the workflow can recover the intended uptake bottleneck. The other three prevent the exercise from becoming a demonstration by construction. In particular, better prediction of conditional repair doesn't entail better intervention choice, and a true but poorly measured decomposition can be less useful than its simpler comparator. Differences in final-accuracy loss are much smaller because initially correct answers dominate that aggregate, which is why the article treats repair conditional on error and harmful intervention as the more diagnostic quantities. The repair

| Constructed world                               | $\Delta$ repair loss | $\Delta$ regret | Result                                                                                |
|-------------------------------------------------|----------------------|-----------------|---------------------------------------------------------------------------------------|
| Uptake bottleneck,<br>accurately measured       | -0.00639             | -0.00657        | The decomposition predicts repair and selects the intervention better.                |
| Conditional signal is<br>sufficient             | +0.05695             | +0.00348        | Conditional coverage wins; the proposed attribution and uptake structure is mistaken. |
| True route structure, noisy<br>feature measures | +0.01478             | +0.00409        | Conditional coverage wins in practice despite the latent structural process.          |
| True route structure, no<br>decision surplus    | -0.00415             | +0.00001        | The decomposition predicts repair better but doesn't improve intervention choice.     |

Table 2: Paired fake-data comparisons. Differences are decomposition minus conditional coverage, so lower logarithmic loss and lower regret appear as negative values.

and regret comparisons in each of the first three worlds have the same direction in all 200 replications; in the fourth, decision regret is effectively tied. The reproducible R script and machine-readable results accompany this resource.

### 3 A CROSSED ROUTE-BY-TASK DESIGN

The worked example uses clinical questions. One item set asks for recommendations recoverable from guidelines or institutional records. A paired set uses the same disease area and answer format and requires a patient-specific lab value, observed sign, dose calculation, or intervention-sensitive causal judgment. The prediction is a retrieval gain on the record-recoverable set and little transfer to the patient-specific set unless the needed measurement or tool route is supplied. Retrieval producing the same gain on both sets would count against the proposed distinction between routes. The design parallels validity argumentation in measurement: inferences from performance are licensed relative to a purpose and tested by varying method while holding the target construct fixed (Kane, 2013; Messick, 1989).

Matching the sets on difficulty cues isn't sufficient. An intervention that supplies a missing lab value also makes the item easier, so a difficulty explanation and a route explanation both predict a gain. Equating measured baseline accuracy weakens the simplest difficulty account without eliminating differences in discrimination, ceiling structure, response variance, latent subskills, or treatment-effect heterogeneity. Matching on noisy estimates also invites regression to the mean. Use baseline performance as a covariate and design diagnostic, not as evidence that difficulty has been removed.

The primary design is crossed rather than merely matched. Build item families in which the relevant route can be supplied or withheld within closely related items, then cross route availability against task requirement. Estimate the interaction in a hierarchical model with varying effects for items, domains, prompt templates, and systems. Declare in advance the smallest interaction that would matter, report its interval, and run reasonable alternatives for item matching, scoring, and exclusion rather than relying on one sample.

In the single-pair implementation, the comparison uses two named chat arrangements that differ only in whether retrieval over a fixed corpus is available, with the prompt template held constant. The

estimate is conditional on that pair; variation across model families and deployment setups lies outside the estimand and belongs to a multi-system extension. The population is a clinical item set split into record-recoverable and patient-specific halves. Raters classify the items from their text without seeing system output, and the classification is frozen before the first run.

The primary outcome is substantive correctness of the recommendation, scored blind, which applies to both halves of the item set. Whether verbatim anchors resolve is secondary and applies only within the record-dependent half. Making anchor resolution primary would manufacture the interaction because patient-specific items have no corpus anchors to resolve.

The worked prediction is that the retrieval gain on record-recoverable items exceeds the gain on patient-specific items by at least ten percentage points. Ten is illustrative, not derived. A defensible threshold needs a decision basis: the cost of a wrong recommendation, the cost of retrieval latency, and what a deployment would do differently at nine points rather than eleven. Fix the value before outcomes are observed so that the prediction can fail.

The interval is interpreted against that threshold rather than zero. Concentration above ten points supports the worked claim; concentration near zero or on the wrong side counts against it. An interval extending from below zero to well above ten describes an uninformative experiment, not evidence either way. The threshold isn't revised after the effect becomes visible.

A stronger implementation crosses both interventions with both task types. Retrieval should selectively help record-dependent items; a measurement tool should selectively help items requiring measurement. This double dissociation tests the proposed route carving, though a relevance-sensitive information account can predict the same qualitative interaction.

#### 4 CHECKING-ROUTE INDEPENDENCE

Compare two checkers: one from the generator's lineage and the other trained and calibrated on separate data and feedback. Give them the same candidate output and task evidence supplied from outside the checkers. Match standalone accuracy, confidence, latency, response format, and revision policy as closely as possible. Measure detection and correction conditional on generator error, especially under perturbations expected to produce common-mode failure.

Verify rather than assume the match. Use held-out calibration items to compare standalone accuracy, confidence calibration, and latency, and route both checker outputs through the same response format and revision policy. The primary cross-route quantity is conditional error overlap: how often the checker repeats the generator's error when the generator is wrong. Report it together with false interventions on generator outputs that were already correct.

For this comparison, input coverage is measured before either checker produces a task-specific signal. Checker outputs are downstream consequences of the architecture, not part of baseline input coverage. If coverage is measured after a checker responds, the predicted advantage is already a conditional-coverage effect.

The decomposition predicts an advantage for the checker with lower conditional error overlap. Calibration history matters only insofar as it predicts that overlap; if admissible perturbations produce no difference, the decomposition has no reason to privilege one history. An input-coverage account predicts no advantage once externally supplied task information and standalone accuracy are matched. A conditional-coverage account can predict one by representing the joint error structure, but that prediction incorporates the relation that checking-route independence names.

## 5 A TEMPORAL TEST OF LIVE ANSWERABILITY

Give two arrangements the same target information at the outset, one holding a frozen pasted record and the other holding a live route to its source. In the update arm, correct the source record and ask a follow-up whose answer turns on the correction. In the control arm, leave the record unchanged. The live route should track the update in the first arm and match the frozen copy in the second. Figure S1 summarizes the design.

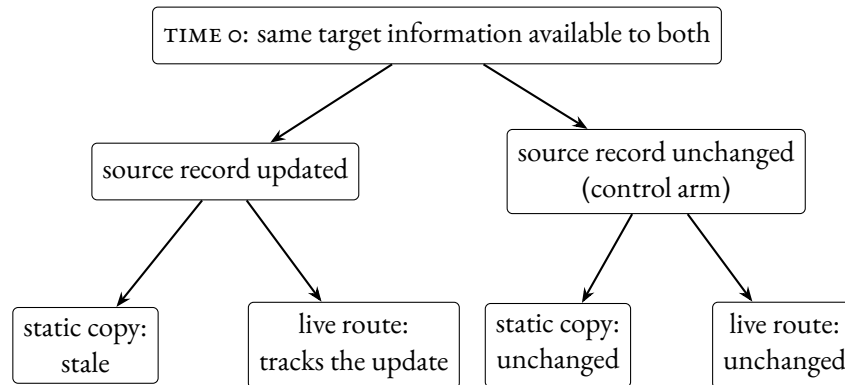


Figure S1: A temporal test of live answerability. At time 0, both arrangements have the same target information. In the update arm, the source record is corrected: the static copy stays stale while the live route transmits the correction. In the control arm, neither should change. No outcome magnitudes are implied.

A dynamic coverage account predicts the same pattern because the arrangements receive different information at follow-up. The manipulation operationalizes live answerability and isolates updateability from initial content equivalence; it doesn't adjudicate between the decomposition and dynamic coverage.

## 6 MANIPULATION CHECKS AND FAILURE CONDITIONS

Failure of a declared route-specific interaction or update effect counts against the decomposition only after confirming that the focal route was invoked and that its output reached production in time to affect the answer. A stale index, failed tool call, or ignored checker signal shows that the intended manipulation wasn't instantiated. Once those checks pass, failure of the declared contrast is evidence that the proposed route carving or corrective feature has been misdescribed.

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